

Problem 19. A population of birds is foraging on a group of three islands. Each morning, all of the birds on each island fly to a different island, dividing themselves evenly between the other two islands.

- Let $x(n)$, $y(n)$, and $z(n)$ represent the number of birds on each island on day n . Find a transition matrix that expresses $x(n+1)$, $y(n+1)$, and $z(n+1)$ as a function of $x(n)$, $y(n)$ and $z(n)$.
- Find the eigenvectors and eigenvalues of the transition matrix.
- If on the first day, the whole population of 900 birds is on one of the islands, explain how the birds will be distributed across the three islands after many days have passed.

Solution.

- Each day, island x receives half the birds previously on island y and half the birds previously on island z , and likewise for the other two islands. Thus

$$x(n+1) = y(n)/2 + z(n)/2$$

$$y(n+1) = x(n)/2 + z(n)/2$$

$$z(n+1) = x(n)/2 + y(n)/2$$

and so

$$\begin{bmatrix} x(n+1) \\ y(n+1) \\ z(n+1) \end{bmatrix} = \begin{bmatrix} 0 & 1/2 & 1/2 \\ 1/2 & 0 & 1/2 \\ 1/2 & 1/2 & 0 \end{bmatrix} \begin{bmatrix} x(n) \\ y(n) \\ z(n) \end{bmatrix}$$

For future use, I will call the transition matrix A , and the state vector v_n , so the above is $v_{n+1} = Av_n$.

- To find the eigenvalues of A , we compute the characteristic polynomial:

$$\det \begin{bmatrix} -t & 1/2 & 1/2 \\ 1/2 & -t & 1/2 \\ 1/2 & 1/2 & -t \end{bmatrix} = -t^3 + 1/8 + 1/8 - (-t)(1/4) - (-t)(1/4) - (-t)(1/4) = -t^3 + 3/4t + 1/4$$

This factors as $(1-t)(-1/2-t)^2$ and thus the eigenvalues are 1 and $-1/2$.

The eigenspace V_1 is

$$V_1 = \ker(A - I_3) = \ker \begin{bmatrix} -1 & 1/2 & 1/2 \\ 1/2 & -1 & 1/2 \\ 1/2 & 1/2 & -1 \end{bmatrix}$$

Adding the first and second row to the third, this becomes

$$\begin{bmatrix} -1 & 1/2 & 1/2 \\ 1/2 & -1 & 1/2 \\ 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -1/2 & -1/2 \\ 1/2 & -1 & 1/2 \\ 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -1/2 & -1/2 \\ 0 & -3/4 & 3/4 \\ 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & -1/2 & -1/2 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix} \rightsquigarrow \begin{bmatrix} 1 & 0 & -1 \\ 0 & 1 & -1 \\ 0 & 0 & 0 \end{bmatrix}$$

and thus $x = z$, $y = z$, $V_1 = \text{span} \left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right\}$ The other eigenspace is

$$V_{-1/2} = \ker \begin{bmatrix} 1/2 & 1/2 & 1/2 \\ 1/2 & 1/2 & 1/2 \\ 1/2 & 1/2 & 1/2 \end{bmatrix}$$

The reduced row echelon form of this is

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

and the kernel, $V_{-1/2}$, is $\text{span} \left\{ \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix} \right\}$.

For future use, we name the eigenvectors as follows:

$$u = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, v = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, w = \begin{bmatrix} 1 \\ 0 \\ -1 \end{bmatrix}$$

(c) If on the first day, all the birds are on island x , then we have

$$v_0 = \begin{bmatrix} 900 \\ 0 \\ 0 \end{bmatrix}$$

We would like to understand $v_n = A^n v_0$ for large n . To do this, we will write v_0 as a linear combination of the eigenvectors of A we found in part (b). This comes down to solving the linear system with augmented matrix

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 900 \\ 1 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 \end{array} \right]$$

The row operations are as follows, starting by adding the second and third rows to the first:

$$\left[\begin{array}{ccc|c} 3 & 0 & 0 & 900 \\ 1 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 300 \\ 1 & -1 & 0 & 0 \\ 1 & 0 & -1 & 0 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 300 \\ 0 & -1 & 0 & -300 \\ 0 & 0 & -1 & -300 \end{array} \right] \rightsquigarrow \left[\begin{array}{ccc|c} 1 & 0 & 0 & 300 \\ 0 & 1 & 0 & 300 \\ 0 & 0 & 1 & 300 \end{array} \right]$$

Thus we have $v_0 = 300u + 300v + 300w$. Using the fact that $A^n u = 1^n u = u$, $A^n v = (-1/2)^n v$, and $A^n w = (-1/2)^n w$, we have

$$v_n = A^n(300u + 300v + 300w) = 300u + 300(-1/2)^n v + 300(-1/2)^n w$$

As $n \rightarrow \infty$, $(-1/2)^n \rightarrow 0$, and so $v_n \rightarrow 300u$. Thus for large n , we expect v_n to be close to

$$300u = \begin{bmatrix} 300 \\ 300 \\ 300 \end{bmatrix}$$

This is to say, the birds are approximately evenly distributed, with 300 on each island.